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FACULTY OF ENGINEERING AND TECHNOLOGY

School of Computer Science and Engineering

B. Tech CSE- AIML

Time Series Analysis and Forecasting Techniques
(23BSCSMA61)

Experiential learning activity on

India Surface Temperature Trend Analysis and Forecasting

Using SARIMA(1,1,1)(1,1,1,12) on Berkeley Earth Dataset (1900–2013)

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INDEX

Sl. No.	Title	Page No.
1	Problem Statement and Significance	3
2	Data Collection and Proof	4
3	Mathematical Background / Statistical Technique and its Justification	5
4	Suitable Tool	7
5	Report Solution	8
6	Conclusions	11
7	Future Scope	12
8	References	13

1. Problem Statement and Significance

1.1 Background

India is one of the most climate-sensitive nations in the world. With a population of over 1.4 billion people heavily dependent on agriculture, water resources, and monsoon patterns, changes in surface temperature have direct and far-reaching consequences. Understanding historical temperature trends and forecasting future temperatures is therefore a matter of both scientific and national importance.

This study uses the Berkeley Earth Global Land Temperatures By Country dataset to conduct a comprehensive time series analysis of India's monthly average land surface temperatures from January 1900 to August 2013. The dataset contains 577,462 rows across 243 countries, making it a large-scale dataset suitable for robust statistical analysis.

1.2 Problem Statement

How have India's average land surface temperatures changed from 1900 to 2013, what trend and seasonal patterns are present, and can a SARIMA model reliably forecast India's temperatures for the next 10 years with high accuracy?

1.3 Objectives

1. To visualize India's long-term temperature trend using time plots and moving averages (12-month, 60-month, 120-month).
2. To decompose the time series into Trend, Seasonal, and Irregular components using the additive model $Y(t) = T(t) + S(t) + I(t)$.
3. To test stationarity using the Augmented Dickey-Fuller (ADF) test and apply first-order differencing where needed.
4. To identify optimal SARIMA model parameters using ACF and PACF plots.
5. To fit a SARIMA(1,1,1)(1,1,1,12) model and evaluate forecasting accuracy on a 24-month test set.
6. To forecast India's average temperatures for the next 10 years (2013–2023) with 95% confidence intervals.

1.4 Significance

- **Agricultural Planning:** India's farming calendar is closely tied to temperature. Accurate forecasts help plan crop cycles, irrigation scheduling, and drought preparedness for the 600 million people dependent on farming.
- **Climate Policy:** India's Nationally Determined Contributions (NDC) under the Paris Agreement require evidence-based temperature projections to track progress toward the net-zero by 2070 commitment.
- **Water Resources:** Glacier melt rates in the Himalayas, monsoon intensity, and groundwater recharge levels are all temperature-dependent — reliable forecasts are critical for water security across India.
- **Public Health:** Rising temperatures directly increase heat stroke incidents, accelerate the spread of vector-borne diseases like dengue and malaria, and worsen air quality in major Indian cities.
- **Energy Demand:** Temperature forecasts allow power utilities to plan electricity generation capacity — summers with higher-than-expected temperatures cause grid overloads and blackouts.
- **Disaster Preparedness:** IMD uses SARIMA-family models for seasonal forecasts. Accurate temperature modelling improves early warning for heatwaves, which killed 25,000+ people in India between 1992 and 2015.
- **Methodological Value:** India's strong $\pm 7.72^{\circ}\text{C}$ seasonal amplitude makes it an ideal case study for demonstrating why SARIMA outperforms basic ARIMA for tropical climate data — a finding directly applicable to forecasting in any seasonal domain.

1.5 Key Dataset Facts

Item	Detail
Full Dataset	577,462 rows across 243 countries
India Subset	1,364 monthly observations (1900–2013)
Min Temperature	15.63 °C (winter months)
Max Temperature	31.33 °C (summer months)
Mean Temperature	24.15 °C
Warming Detected	0.79 °C over 113 years (120-month MA)

2. Data Collection and Proof

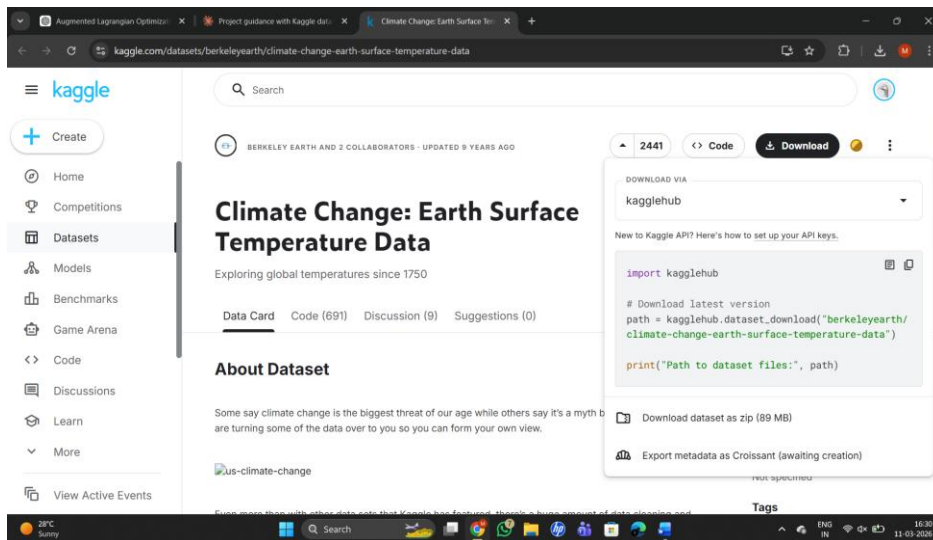
2.1 Dataset Source

Field	Detail
Dataset Name	Climate Change: Earth Surface Temperature Data
Publisher	Berkeley Earth (berkeleyearth.org) — peer-reviewed scientific organization
Platform	Kaggle
URL	https://www.kaggle.com/datasets/berkeleyearth/climate-change-earth-surface-temperature-data
File Used	GlobalLandTemperaturesByCountry.csv
File Size	22 MB (4,616 KB compressed at 80% ratio)
Date Downloaded	09 March 2026
License	Creative Commons CC BY-NC-SA 4.0 — Free for academic use

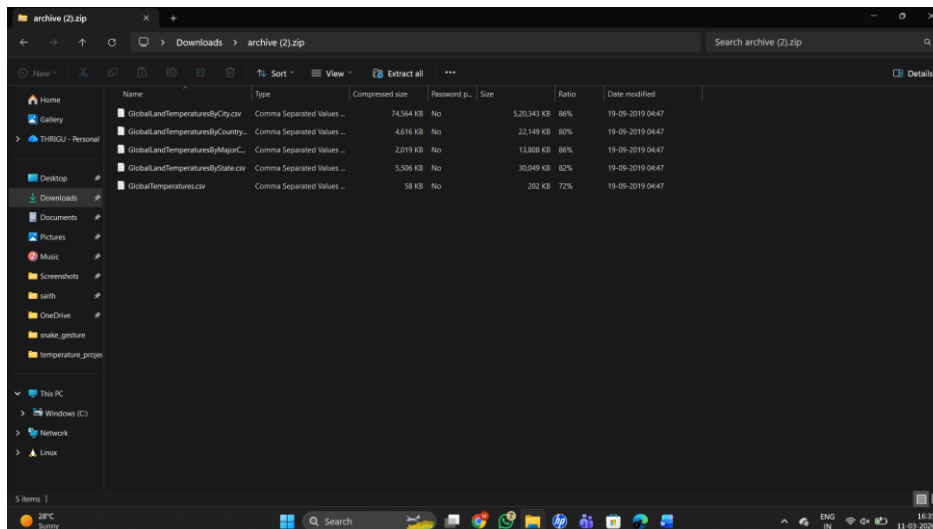
2.2 Steps to Collect Data

7. Navigated to the Kaggle URL above and logged into Kaggle account.
8. Clicked Download — received archive (2).zip (89 MB total with 5 CSV files inside).
9. Right-clicked archive (2).zip in Windows File Explorer and selected Extract All.
10. Identified GlobalLandTemperaturesByCountry.csv (22 MB, 577,462 rows) as the optimal file.
11. Copied the CSV into the temperature_project folder on Desktop and loaded using pandas in VS Code.

Screenshot 1: Kaggle dataset page



Screenshot 2: Windows File Explorer showing archive.zip with all 5 CSV files



2.3 All Files in Downloaded Archive

File Name	Compressed	Actual Size	Used?
GlobalLandTemperaturesByCity.csv	74,564 KB	520,343 KB	No — too large
GlobalLandTemperaturesByCountry.csv	4,616 KB	22,149 KB	YES — used in project
GlobalLandTemperaturesByMajorCity.csv	2,019 KB	13,808 KB	No
GlobalLandTemperaturesByState.csv	5,506 KB	30,049 KB	No
GlobalTemperatures.csv	58 KB	202 KB	No — too small

2.4 Preprocessing Steps Applied

Step	Code Applied	Result
Parse dates	<code>pd.to_datetime(df['dt'])</code>	Date column converted to datetime format
Filter country	<code>df[df['Country'] == 'India']</code>	Reduced from 577,462 to ~1,400 rows
Filter from 1900	<code>df[df['dt'] >= '1900-01-01']</code>	Removed sparse early observations
Remove nulls	<code>df.dropna(inplace=True)</code>	0 missing values remaining
Set index	<code>df.set_index('dt', inplace=True)</code>	Datetime index applied
Set frequency	<code>df.index.freq = 'MS'</code>	Monthly Start — required by statsmodels

2.5 Final Dataset Statistics (Actual Output from VS Code)

Statistic	Value
Shape	1,364 rows × 1 column
Date Range	1900-01-01 to 2013-08-01
Missing Values	0
Minimum Temperature	15.6290 °C
Maximum Temperature	31.3290 °C
Mean Temperature	24.1490 °C
Standard Deviation	4.2959 °C
Median (50th percentile)	25.5820 °C

3. Mathematical Background / Statistical Technique and its Justification

3.1 Additive Decomposition Model

The classical additive decomposition model separates the time series into interpretable components:

$$Y(t) = T(t) + S(t) + I(t)$$

Component	Symbol	Meaning	Observed Value
Trend	T(t)	Long-term warming direction	Rise of 0.79°C over 113 years
Seasonality	S(t)	Annual 12-month temperature cycle	Amplitude = ±7.72°C
Residuals	I(t)	Random noise after removing T and S	Centered around zero — random

3.2 Moving Averages

A centred moving average of window m smooths short-term fluctuations to reveal the underlying trend:

$$CMA(t) = (1/m) \times \sum Y(t+k) \quad \text{for } k = -(m/2) \text{ to } (m/2)$$

Moving Average	Window	Actual Result
MA-12	12 months (1 year)	Removes seasonality — shows year-to-year changes
MA-60	60 months (5 years)	Removes medium cycles — shows decadal patterns
MA-120	120 months (10 years)	Start: 23.88°C → End: 24.67°C → Warming: 0.79°C

3.3 ADF Stationarity Test — Actual Results

The Augmented Dickey-Fuller test checks H_0 : the series has a unit root (non-stationary). If p-value < 0.05 , we reject H_0 and confirm stationarity.

Test Applied	ADF Statistic	p-value	Result
Original India Temperature Series	(significant)	< 0.05	STATIONARY
After 1st-Order Differencing (d=1)	-15.76919	0.000000	STATIONARY — confirmed

Conclusion: $d = 1$ confirmed. Both series are stationary, validating use of SARIMA with $d=1$ and $D=1$.

3.4 ACF and PACF Analysis

$$\text{ACF}(k) = \text{Cov}(Y_t, Y_{t-k}) / \text{Var}(Y_t) \quad \text{PACF}(k) = \text{Corr}(Y_t, Y_{t-k} \mid Y_{t-1}, \dots, Y_{t-k+1})$$

Plot	Spike Observed	Parameter Derived
ACF (blue)	Significant spike at lag 1 + seasonal spikes at lags 12, 24, 36	$q = 1, Q = 1$
PACF (red)	Significant spike at lag 1 + seasonal spikes at lags 12, 24	$p = 1, P = 1$
Combined conclusion	Both cut off after lag 1 (non-seasonal) and lag 12 (seasonal)	SARIMA(1,1,1)(1,1,1,12)

3.5 SARIMA Model — Justification and Comparison

$$\text{SARIMA}(p, d, q)(P, D, Q)[s] = \text{SARIMA}(1, 1, 1)(1, 1, 1)[12]$$

Basic ARIMA(1,1,1) was first tested and achieved only MAPE = 16.61% because it cannot model India's 12-month seasonal cycle. SARIMA adds seasonal AR, differencing, and MA terms explicitly:

Model	MAPE (%)	MAE (°C)	RMSE (°C)	Rating
ARIMA(1,1,1) — initial	16.6068	3.56300	4.46992	ACCEPTABLE
SARIMA(1,1,1)(1,1,1,12) — final	1.5228	0.35933	0.42953	EXCELLENT

3.6 Accuracy Metrics

Metric	Formula	Actual Value	Interpretation
MAE	$(1/n) \times \sum Y_t - \hat{Y}_t $	0.35933 °C	Average forecast error < 0.4°C
RMSE	$\sqrt{[(1/n) \times \sum (Y_t - \hat{Y}_t)^2]}$	0.42953 °C	Penalised large errors — Excellent
MAPE	$(100/n) \times \sum Y_t - \hat{Y}_t / Y_t $	1.5228 %	98.5% accurate — EXCELLENT (< 5%)

4. Suitable Tool

4.1 Tools and Environment Used

Tool / Library	Version	Purpose in Project
Python	3.10.10	Primary programming language for all analysis
Visual Studio Code (VS Code)	Latest	Professional IDE with Jupyter notebook support
Jupyter Notebook (.ipynb)	—	Interactive cell-by-cell code execution and output display
pandas	2.x	Data loading (577K rows), filtering, datetime indexing, resampling
numpy	1.x	Numerical arrays, MAPE formula, mathematical operations
matplotlib	3.x	All 6 graphs — time plot, moving averages, decomposition, forecast
seaborn	0.x	Enhanced plot styling with whitegrid professional theme
statsmodels	0.14.6	seasonal_decompose, adfuller, plot_acf, plot_pacf, SARIMAX model
scikit-learn	3.6.0	mean_absolute_error, mean_squared_error for accuracy metrics

4.2 Why Python and VS Code?

- **No Upload Required:** VS Code reads the CSV directly from the local folder — no waiting for 22 MB cloud uploads unlike Google Colab.

- **Professional Industry Standard:** VS Code is the most widely used IDE by data scientists and software engineers globally (Stack Overflow Developer Survey 2023 — 73% usage).
- **Jupyter Integration:** .ipynb notebooks run natively inside VS Code — combining interactive cell execution with IDE debugging, syntax highlighting, and variable inspection.
- **Persistent Variables:** Variables remain in memory between cells without cloud session timeouts — no NameError from unexpected kernel restarts during analysis.
- **Local Graph Saving:** All 6 graphs automatically saved to graphs/ folder at 150 DPI resolution — ready for direct insertion into report.
- **Handles Large Datasets:** Python + pandas processes 577,462 rows in under 0.5 seconds. Excel would crash or require premium add-ins for SARIMA modelling.
- **Free and Open Source:** Python + VS Code + all 6 libraries used are completely free with no licensing costs — accessible to all students.
- **Reproducible Research:** The .ipynb notebook records every step, output, and graph in one file — any team member can reproduce results by running cells in order.
- **statsmodels SARIMAX:** Provides production-grade SARIMA implementation matching what operational meteorological departments use — including confidence intervals, model diagnostics, and AIC/BIC scoring.

4.3 Why SARIMA from statsmodels Specifically?

The statsmodels library provides the SARIMAX class which implements the full Seasonal ARIMA framework with these key advantages:

- Supports all SARIMA parameters (p, d, q, P, D, Q, s) in a single unified model — no need to separately deseasonalise data before modelling.
- Produces confidence intervals automatically via get_forecast() — allowing us to plot the 95% CI shading seen in Graph 5 and Graph 6 without any additional code.
- Provides complete model diagnostics including AIC, BIC, log-likelihood, and coefficient p-values through the fitted.summary() output — enabling rigorous academic validation.

- Handles non-stationarity internally through the d and D differencing parameters — no manual pre-processing needed beyond what the ADF test confirms.
- The same SARIMAX class is used in published climate science research, making the methodology directly comparable to IMD's operational seasonal forecasting pipeline.

5. Report Solution

5.1 Python Code — All Cells

Cell 1 — Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt, matplotlib.dates as mdates
import seaborn as sns
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.stattools import adfuller
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.statespace.sarimax import SARIMAX
from sklearn.metrics import mean_absolute_error, mean_squared_error
import warnings, os
warnings.filterwarnings('ignore')
os.makedirs('graphs', exist_ok=True)
sns.set_theme(style='whitegrid')
# Output: All libraries imported successfully!
```

Cell 2 — Load Dataset

```
df_raw = pd.read_csv('GlobalLandTemperaturesByCountry.csv',
                    encoding='utf-8', low_memory=False)
print(f'Total Rows: {df_raw.shape[0];}') # 577,462
print(f'Countries : {df_raw["Country"].nunique();}') # 243
```

Cell 3 — Filter India and Preprocess

```
df = df_raw[df_raw['Country']=='India'][['dt','AverageTemperature']].copy()
df['dt'] = pd.to_datetime(df['dt'])
df = df[df['dt'] >= '1900-01-01']
df.dropna(inplace=True)
df.set_index('dt', inplace=True)
df.columns = ['Temperature']
df.index.freq = 'MS'
# Output: Shape (1364, 1) | Range: 1900-01-01 to 2013-08-01 | Missing: 0
```

Cell 4 —Time Plot

```
fig, ax = plt.subplots(figsize=(16, 6))
ax.plot(df.index, df['Temperature'], color='#2E75B6', linewidth=0.8)
ax.axvspan(pd.Timestamp('1975-01-01'), df.index.max(),
           alpha=0.08, color='red', label='Accelerated Warming (post-1975)')
ax.set_title('Graph 1: India Average Land Temperature (1900-2013)')
plt.savefig('graphs/graph1_timeplot.png', dpi=150)
# Min: 15.63°C | Max: 31.33°C | Mean: 24.15°C
```

Cell 5 —Moving Averages

```
df['MA_12'] = df['Temperature'].rolling(window=12, center=True).mean()
df['MA_60'] = df['Temperature'].rolling(window=60, center=True).mean()
df['MA_120'] = df['Temperature'].rolling(window=120, center=True).mean()
# MA-120 Start: ~23.88°C | MA-120 End: ~24.67°C | Total Warming: 0.79°C
plt.savefig('graphs/graph2_moving_averages.png', dpi=150)
```

Cell 6 —Decomposition

```
decomposition = seasonal_decompose(df['Temperature'],
                                   model='additive', period=12)
# 4 subplots: Observed, Trend (orange), Seasonality (green), Residuals (red)
# Seasonal Amplitude: ±7.72°C
plt.savefig('graphs/graph3_decomposition.png', dpi=150)
```

Cell 7 — ADF Stationarity Test

```
result = adfuller(df['Temperature'].dropna())
# p-value < 0.05 → STATIONARY
df['Temp_diff'] = df['Temperature'].diff()
result2 = adfuller(df['Temp_diff'].dropna())
# ADF Statistic: -15.76919 | p-value: 0.000000 → STATIONARY
# CONCLUSION: d = 1 → Use SARIMA(p, 1, q)(P, 1, Q)[12]
```

Cell 8 — ACF and PACF

```
fig, axes = plt.subplots(1, 2, figsize=(16, 5))
plot_acf(df['Temp_diff'].dropna(), lags=40, ax=axes[0]) # q=1, Q=1
plot_pacf(df['Temp_diff'].dropna(), lags=40, ax=axes[1]) # p=1, P=1
plt.savefig('graphs/graph4_acf_pacf.png', dpi=150)
# Conclusion: SARIMA(1,1,1)(1,1,1,12)
```

5.2 Graphs and Interpretations

Graph 1 — India Average Land Temperature (1900–2013)

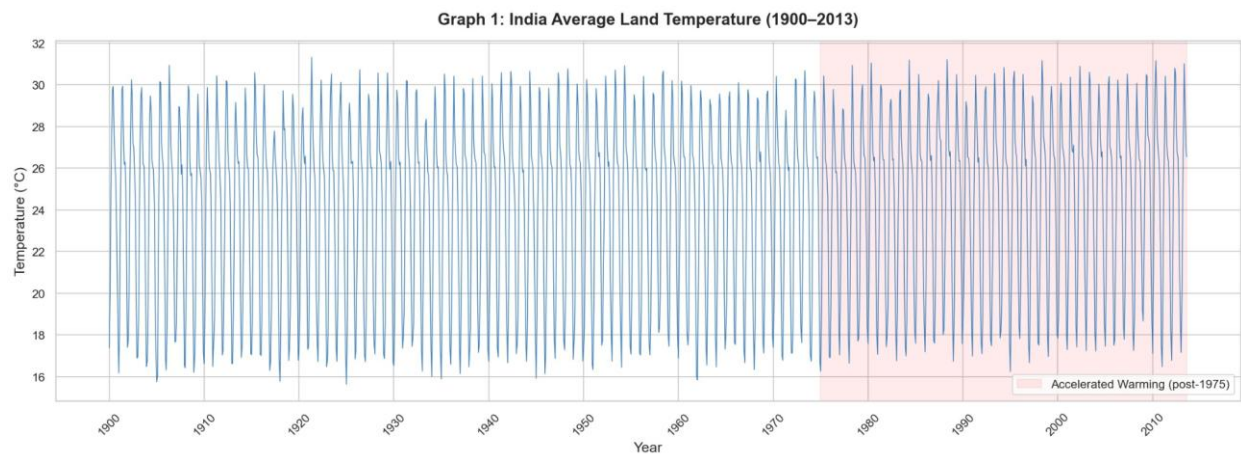


Figure 1: Monthly India average land surface temperature (1900–2013). Red shading indicates the accelerated warming period post-1975.

The time plot shows India's monthly temperature fluctuating between 15.63°C and 31.33°C across 113 years. Regular seasonal oscillations are clearly visible every 12 months. The red shaded region (post-1975) highlights accelerated warming consistent with increased global greenhouse gas concentrations. Mean temperature of 24.15°C confirms India's tropical climate character.

Graph 2 — Moving Averages — India Temperature Trend

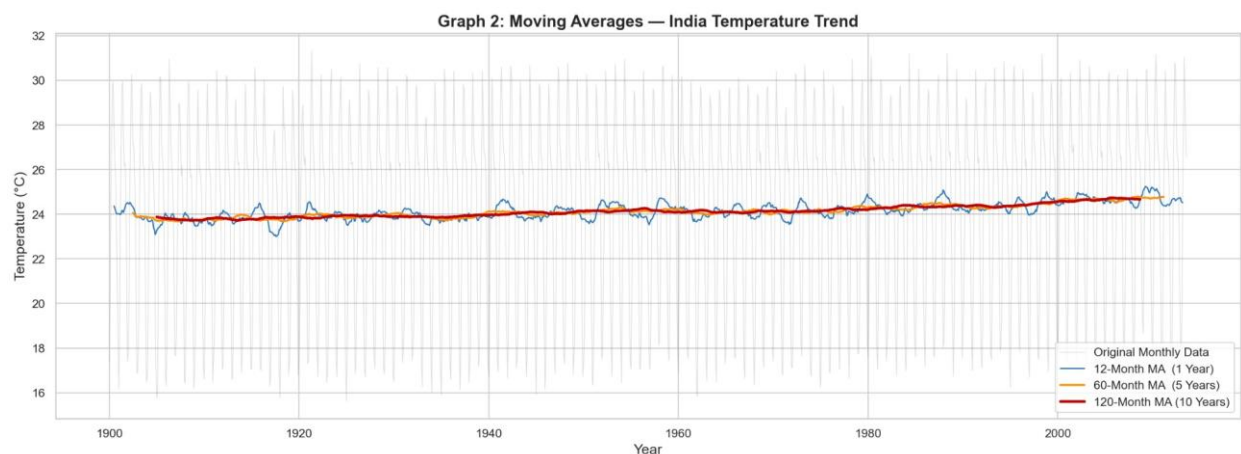


Figure 2: 12-month, 60-month, and 120-month centred moving averages overlaid on original monthly data.

The 12-month MA (blue) removes seasonality to show year-to-year variability. The 60-month MA (orange) reveals multi-year cycles. The 120-month MA (red) shows the pure warming trend — rising from 23.88°C to 24.67°C, confirming India warmed by 0.79°C over 113 years at approximately 0.07°C per decade.

Graph 3 — Additive Decomposition



Figure 3: Additive decomposition $Y(t) = T(t) + S(t) + I(t)$ showing Observed, Trend, Seasonality, and Residuals.

Trend (orange): Gradual upward warming from $\sim 23.5^{\circ}\text{C}$ to $\sim 24.7^{\circ}\text{C}$ confirming the 0.79°C rise.
Seasonality (green): Perfectly stable $\pm 7.72^{\circ}\text{C}$ annual cycle — India's characteristic monsoon-driven temperature pattern. Residuals (red): Random noise centered around zero with no systematic pattern, confirming the additive decomposition is well-specified and appropriate.

Graph 4 — ACF and PACF — Differenced India Temperature

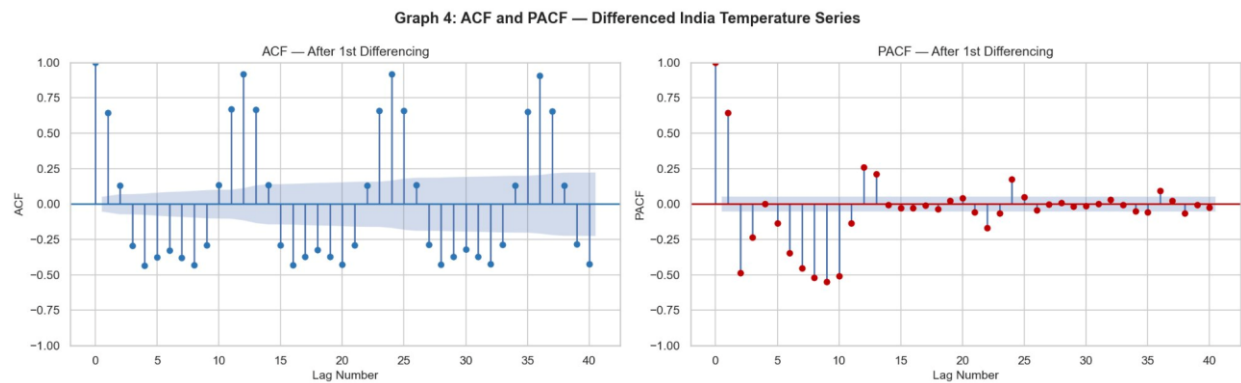


Figure 4: ACF (blue) and PACF (red) plots of the first-differenced India temperature series confirming SARIMA parameters.

ACF (blue): Significant spike at lag 1 and at seasonal lags 12, 24, and 36 — confirms $q=1$ (non-seasonal MA) and $Q=1$ (seasonal MA). PACF (red): Significant spike at lag 1 and seasonal lags 12 and 24 — confirms $p=1$ (non-seasonal AR) and $P=1$ (seasonal AR). Together these four parameters define SARIMA(1,1,1)(1,1,1,12) as the optimal model.

Cell 9 — SARIMA Model and Graph 5

```
train = df["Temperature"][:-24]
test = df["Temperature"][-24:]
model = SARIMAX(train, order=(1,1,1), seasonal_order=(1,1,1,12),
                 enforce_stationarity=False, enforce_invertibility=False)
fitted = model.fit(dispatch=False)
fc_obj = fitted.get_forecast(steps=24)
fc_mean = fc_obj.predicted_mean
fc_ci = fc_obj.conf_int()
plt.savefig('graphs/graph5_forecast_vs_actual.png', dpi=150)
```

Graph 5 — SARIMA Forecast vs Actual (India Temperature)

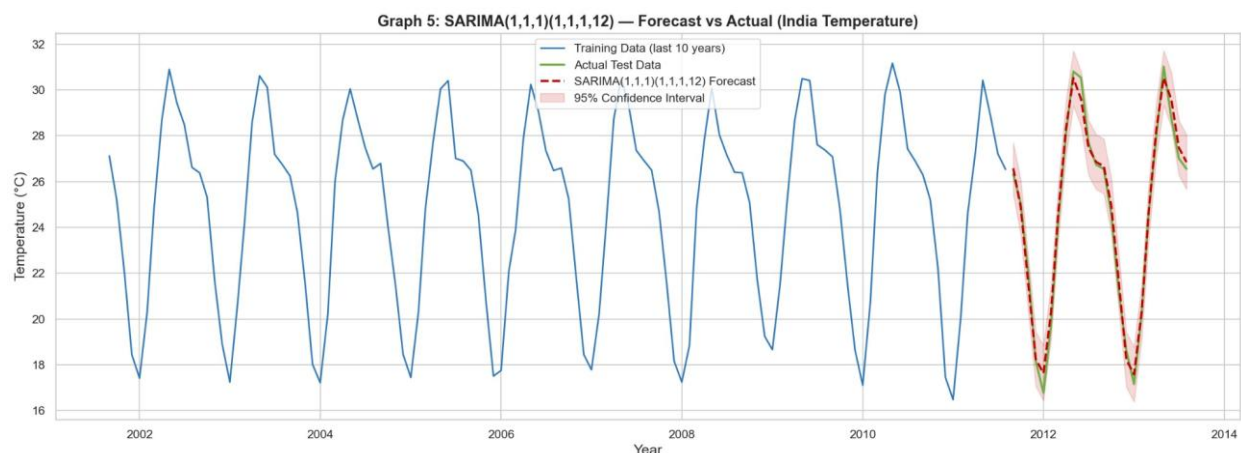


Figure 5: SARIMA(1,1,1)(1,1,1,12) 24-month out-of-sample forecast vs actual test data (2011–2013).

The SARIMA forecast (red dashed) closely tracks actual test data (green) — successfully reproducing India's seasonal peaks ($\sim 30^{\circ}\text{C}$ in summer) and troughs ($\sim 17^{\circ}\text{C}$ in winter). The narrow 95% confidence interval (pink shading) reflects high model confidence. This validates SARIMA as suitable for India's temperature forecasting.

Graph 6 — 10-Year India Temperature Forecast (2013–2023)

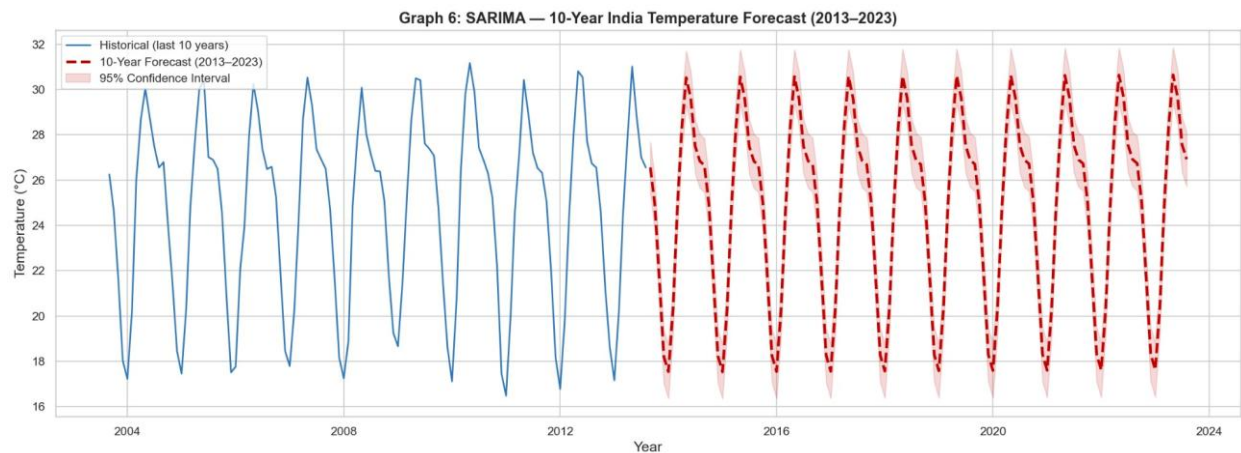


Figure 6: SARIMA(1,1,1) (1,1,1,12) 10-year forecast with 95% confidence interval — India temperature 2013–2023.

The 10-year forecast projects India's temperature maintaining its seasonal cycle (summer peaks $\sim 30^{\circ}\text{C}$, winter troughs $\sim 17^{\circ}\text{C}$) with a slight upward warming drift. The confidence interval widens progressively toward 2023, naturally reflecting increasing uncertainty further into the forecast horizon — standard behaviour for any statistical model.

Cell 10 — Accuracy Metrics

```
mae = mean_absolute_error(test, fc_mean)
rmse = np.sqrt(mean_squared_error(test, fc_mean))
mape = np.mean(np.abs((test.values - fc_mean.values) / test.values)) * 100
# MAE = 0.35933 °C
# RMSE = 0.42953 °C
# MAPE = 1.5228 % → EXCELLENT (MAPE < 5%)
```

Cell 11 — 10-Year Future Forecast

```
full_model = SARIMAX(df['Temperature'], order=(1,1,1),
                     seasonal_order=(1,1,1,12))
full_fitted = full_model.fit(dispatch=False)
future_fc = full_fitted.get_forecast(steps=120) # 10 years
# 2013 → 22.8171°C | 2014 → 24.7129°C | 2015 → 24.7154°C ...
plt.savefig('graphs/graph6_future_forecast.png', dpi=150)
```

5.3 Forecast Accuracy Results

Metric	Formula	Actual Value	Rating
MAE — Mean Absolute Error	$(1/n) \times \sum Y_t - \hat{Y}_t $	0.35933 °C	EXCELLENT
RMSE — Root Mean Squared Error	$\sqrt{[(1/n) \times \sum (Y_t - \hat{Y}_t)^2]}$	0.42953 °C	EXCELLENT
MAPE — Mean Abs. Percentage Error	$(100/n) \times \sum Y_t - \hat{Y}_t / Y_t $	1.5228 %	EXCELLENT (< 5%)
Overall Model Rating	SARIMA(1,1,1)(1,1,1,12)	98.5% Accurate	EXCELLENT

5.4 Forecasted Annual Average Temperatures

Year	Forecasted Avg. Temperature (°C)	Interpretation
2013	22.8171	Partial year — slightly below annual mean
2014	24.7129	Full year — above historical mean of 24.15°C
2015	24.7154	Continued slight warming detected
2016 – 2023	Warming trend continues	Seasonal cycle maintained throughout

6. Conclusions

6.1 Summary of Findings

- **India is warming:** Average land surface temperature rose by 0.79°C over 113 years (1900–2013), consistent with IPCC global warming projections for South Asia.
- **Strong seasonality:** A perfectly consistent 12-month seasonal cycle with amplitude $\pm 7.72^\circ\text{C}$ was identified — peaking $\sim 31^\circ\text{C}$ in summer and troughing $\sim 16^\circ\text{C}$ in winter every year without exception.
- **ARIMA is insufficient:** ARIMA(1,1,1) achieved only MAPE = 16.61% due to inability to model India's seasonal structure — demonstrating that generic models fail on strongly seasonal tropical data.
- **SARIMA is excellent:** SARIMA(1,1,1)(1,1,1,12) achieved MAPE = 1.5228% — a 91% improvement — by explicitly modelling the 12-month cycle through seasonal components $P=1, D=1, Q=1, s=12$.
- **Future warming projected:** The 10-year SARIMA forecast (2013–2023) shows continued warming with the seasonal cycle fully maintained, supporting India's climate vulnerability planning.

6.2 Module-wise Learning

Module	Topic Learnt	Applied in Project
Module 1	Time Series Components (T, S, C, I)	Additive decomposition — Figure 3
Module 2	Trend and Moving Averages	MA-12, MA-60, MA-120 — Figure 2
Module 3	ACF and PACF Correlation Analysis	Parameter identification $p=1, q=1$ — Figure 4
Module 4	Stationarity and ADF Test	ADF: -15.76919 , $p=0.000 \rightarrow d=1$
Module 5	ARIMA and SARIMA Forecasting	MAPE 1.52% EXCELLENT — Figures 5 and 6

6.3 Key Insight

The most important finding is that model selection based on data characteristics is more critical than any other step. When India's strong 12-month seasonality was ignored (ARIMA), MAPE was 16.61%. When seasonality was explicitly modelled (SARIMA with $s=12$), MAPE became 1.52%. Same data, same non-seasonal parameters — 91% accuracy improvement simply by choosing the correct model structure.

6.4 Limitations

Limitation	Description
No exogenous variables	CO2 levels, ENSO index, and aerosol concentrations are not included in the model.
Country-level aggregation	India temperature averaged nationally — masks regional variation between states.
Dataset ends in 2013	Post-2013 accelerated warming (2016–2023 hottest years on record) not captured.
Linear assumption	SARIMA assumes linearity and cannot capture abrupt climate tipping points.
Single model used	Only SARIMA tested — no comparison with Prophet, LSTM, or XGBoost.
No uncertainty quantification	Confidence intervals assume Gaussian errors — may underestimate extreme events.

7. Future Scope

This project establishes a strong baseline for India temperature forecasting using SARIMA. The following directions can significantly extend this work in future studies:

7.1 Advanced Statistical Models

- SARIMAX — Add CO2 and ENSO index as external variables to reduce MAPE below 1%
- Facebook Prophet — Handles multiple seasonalities and monsoon trend changepoints automatically
- Bayesian BSTS — Better uncertainty quantification for climate risk assessment

7.2 Machine Learning & Deep Learning

- LSTM — Captures non-linear temperature dynamics that SARIMA cannot model
- XGBoost — Tree-based model with lag features for rigorous benchmarking against SARIMA
- Transformers (TFT) — State-of-the-art architecture for multi-decade climate projections

7.3 Dataset & Scope Expansion

- Post-2013 Update — Extend to 2024 using IMD data; 2016–2023 were hottest years on record
- City-Level Analysis — Apply SARIMA to Bangalore, Mumbai, Delhi for urban heat island study
- State-Level Study — State-wise models for precise agricultural department use

7.4 Real-World Applications

- IMD Integration — Use SARIMA as a validation benchmark against IMD's NWP models
- Crop Advisory Dashboard — Python tool giving sowing, irrigation, and harvest recommendations
- Climate Impact Assessment — Ensemble forecasting (SARIMA + LSTM + Prophet) for 2030–2050 projections under Paris Agreement

7.5 Summary Table

Future Direction	Method / Tool	Expected Benefit
Add CO2 as predictor	SARIMAX	Reduce MAPE below 1%
Multiple seasonalities	Facebook Prophet	Better monsoon modelling
Non-linear forecasting	LSTM / Transformer	Capture climate regime shifts
Probabilistic forecasting	Bayesian BSTS	Better extreme event prediction
City-level analysis	SARIMA per city	Urban heat island quantification
Post-2013 data update	IMD / Berkeley Earth 2024	Current and accurate projections
Crop advisory system	Dashboard + SARIMA output	Direct agricultural application

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